$= 0.58$ m

23. Ans: A

For the magnetic flux density inside the solenoid, use "$B = \mu_0 n I$"

$$n = \frac{B}{\mu_0 I} = \frac{0.0270}{(4\pi \times 10^{-7})(12.0)} = 1790.5 \text{ turns per metre}$$

Since the solenoid is 40.0 cm long, it contains 716.2 turns (= 1790.5 $\times$ 0.400)

Total length of wire required = 716.2 turns $\times$ circumference of solenoid

$$= 716.2 \times \pi D = 716.2 \times \pi \times 0.028 \text{ m} = 63 \text{ m}$$